Spectral Detection Data Analysis
Homework Report

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Date: February 24, 2026

Executive Summary

This report analyzes RF spectral-snapshot detections produced by an object detector and a tracker. We perform exploratory data analysis (EDA) on a controlled dataset with ground-truth labels and on an uncontrolled operational dataset. In Task 1, we summarize score distributions, signal characteristics (power, SINR, bandwidth), and quantify precision/recall on controlled data as well as a pseudo-FPR behavior on noise-labeled detections in uncontrolled data. In Task 2, we design an alert trigger using a Neyman–Pearson (NP) threshold to satisfy a false-alarm budget and add lightweight reliability gates at the track level (association_id) to reduce false alerts while limiting recall loss. Due to time constraints, the full uncontrolled deep-dive is presented for configuration 1, while configuration 2 is reported at the trigger/summary level.

1. Background and System Overview

The monitored system operates on Radio Frequency (RF) spectral images (spectral snapshots). For each snapshot, a machine-learning object detector outputs candidate signal detections with a confidence score and physical attributes. A tracker then associates detections across consecutive snapshots to form signal tracks, identified by a unique association_id.

Scanning configurations

Two scanning configurations are evaluated, differing by Resolution Bandwidth (RBW). Lower RBW yields finer frequency resolution. Both uncontrolled configurations use a scan rate of 10 Hz.

2. Dataset Description

2.1 Files

• uncontrolled_detections_export_config_1.csv (uncontrolled operational detections; configuration 1)

• uncontrolled_detections_export_config_2.csv (uncontrolled operational detections; configuration 2)

• controlled.csv (controlled experiment with ground-truth labels and RBW field)

2.2 Key columns and definitions

Uncontrolled data include association_id, category, score, start_time/end_time, frequency_low/frequency_high, signal_power (dBm), and sinr (dB). Category 69 is the

legitimate signal of interest in the uncontrolled experiment; category −1 indicates noise/unknown signals and is the main source of false positives. Controlled data include GT labels, Pred Labels, and Pred scores; the GT class 0 represents background noise (no true signal).

2.3 Notation and equations

Signal-to-Interference-plus-Noise Ratio (SINR) is defined as:

Equation (1) $SINR = \frac{S}{I + N}$

where S is signal power, I is interference power, and N is noise power.

For each detection, define center frequency and bandwidth:

Equation (2) $f_c = \frac{(f_{low} + f_{high})}{2}$

Equation (3) $BW = f_{high} - f_{low}$

Binary performance metrics:

Equation (4) $TPR\ (Recall) = \frac{TP}{TP + FN}$

Equation (5) $FPR = \frac{FP}{FP + TN}$

3. Analysis Methodology

All computations are performed in Python using pandas for data handling and matplotlib for visualization. For uncontrolled data, analysis focuses on records from the FINAL layer (post-tracking) when available. Track-level metrics (like average duration per association_id) are computed by grouping detections by association_id.

Figure placeholders

Throughout the report, placeholders indicate recommended figures (histograms, density plots, boxplots, and time-series). Where figures were not generated due to time constraints, numerical summaries are reported instead.

4. Task 1: Recall / Precision and FPR Analysis

The full analysis pipeline was executed for both configurations. To keep the document concise, configuration 1 is presented in detail in this report, while configuration 2 results and all plots/tables can be reproduced from the accompanying Python code

4.1 Statistics

4.1.1 Total number of detections

The total number of detections is computed as the number of rows in each dataframe (df.shape[0]).

```
TASK 1 — CONTROLLED DATASET
================================================================================
Shape: (1161, 4)
Columns: ['GT labels', 'pred labels', 'pred scores', 'rbw']
Total detections: 1161
```

[Table 1: Total detections by dataset/configuration]

4.1.2 Number of unique signals

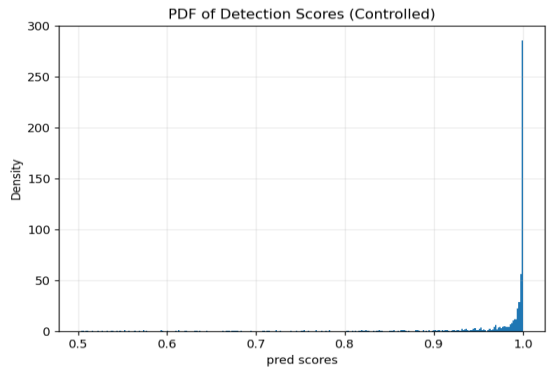
For uncontrolled data, the number of unique tracked signals is computed as nunique(association_id). For controlled data, unique signals can be analyzed per GT label (and optionally per rbw).

```
Unique GT labels (including 0): 13
Unique signals in controlled (excluding 0): 12
```

[Figure 1: Bar chart of unique association_id counts control

4.1.3 Distribution of detection scores

In the controlled dataset, the score distribution is right-skewed with many detections close to 1.0. To avoid bias, score histograms should be interpreted together with predicted labels, because background/no-signal predictions (pred label = 0) are not operational detections and may distort the apparent confidence distribution.



[Figure 2: Score histogram/density plot (controlled)]

4.1.4 Average signal duration per track

Track duration is estimated as (end_time − start_time) and then averaged per association_id; report both per-track mean and the overall distribution to capture variability.

config_1

```
Average duration: 34.60 sec
```

[Figure 3: Distribution of track durations (seconds)]

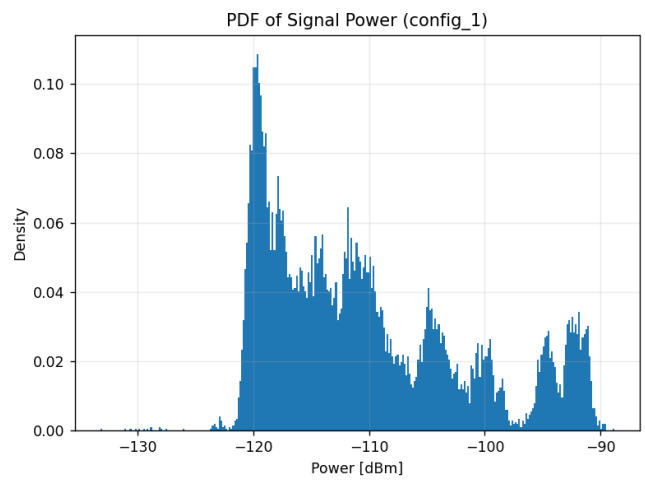
4.2 Signal Characteristics

4.2.1 Signal power distribution

Signal power is summarized using mean/median and visualized using a histogram or boxplot. Power values are reported in dBm (decibel-milliwatts).

Note (units): dBm = 10·log10(P[mW]).

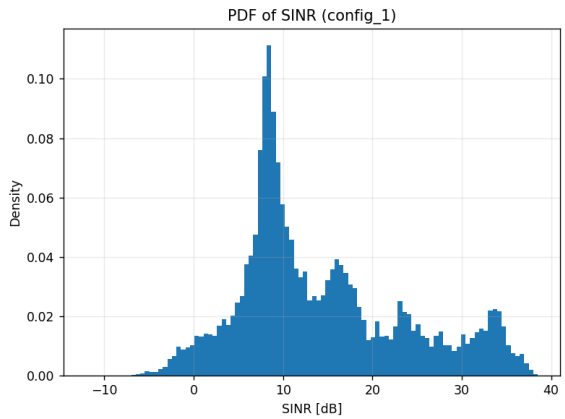
```
Mean signal_power: -110.26 | Median: -112.14
Mean SINR: 14.25 | Median: 11.28
```



[Figure 4: signal power histogram]

4.2.2 SINR distribution

SINR values are visualized and compared across categories. Draft observation: many detections show high-quality signals (SINR well above the noise floor), indicating strong separability in many cases.



[Figure 5: SINR histogram]

4.2.3 Frequency band utilization

Frequency utilization is analyzed by computing f_c and BW (Equations 2–3) and plotting their distributions. Draft observation: the spectrum clusters around discrete center frequencies (e.g., around ~865 MHz), indicating narrowband emitters and discrete channels.

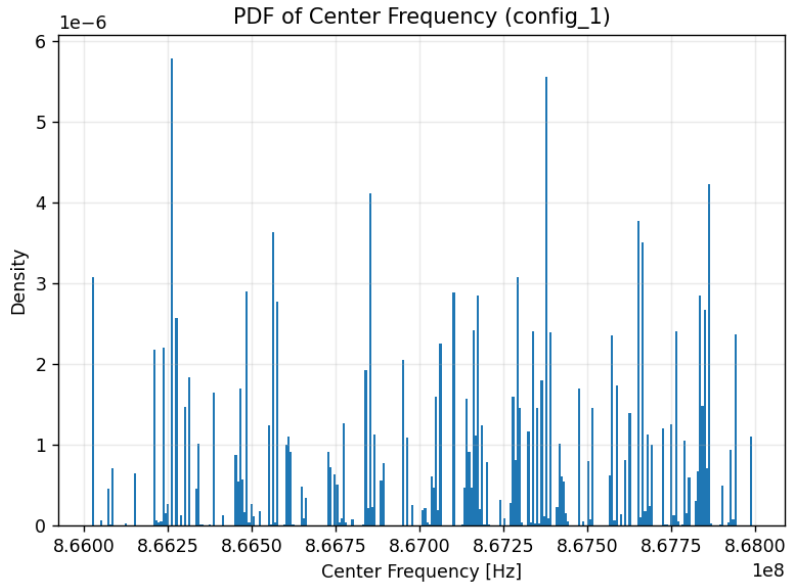
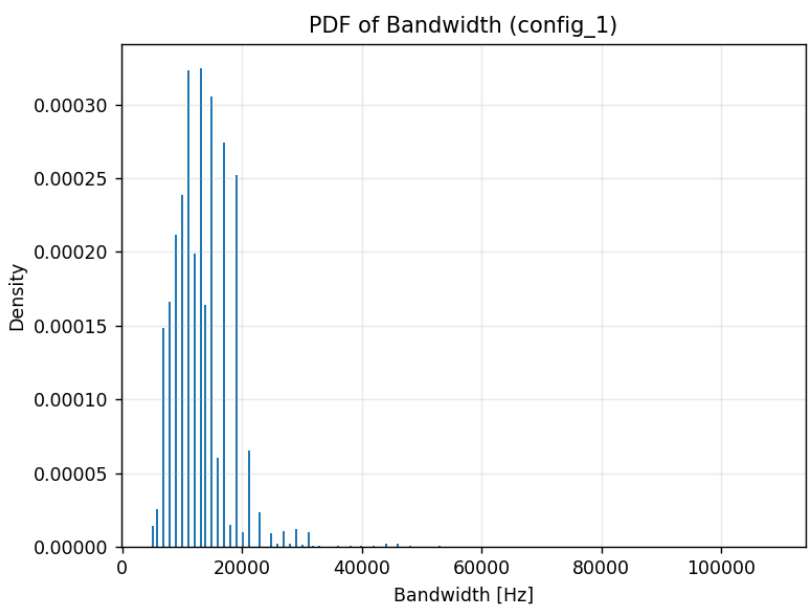


Figure 6: KDE/histogram of center frequency f_c]



[Figure 7: Bandwidth distribution BW]

4.2.4 Temporal coverage pattern

The distribution of detection events over time is examined using event counts per unit time (e.g., per minute). If modeling event arrivals, a Poisson process assumption can be tested by comparing empirical counts to a Poisson model λ(t).

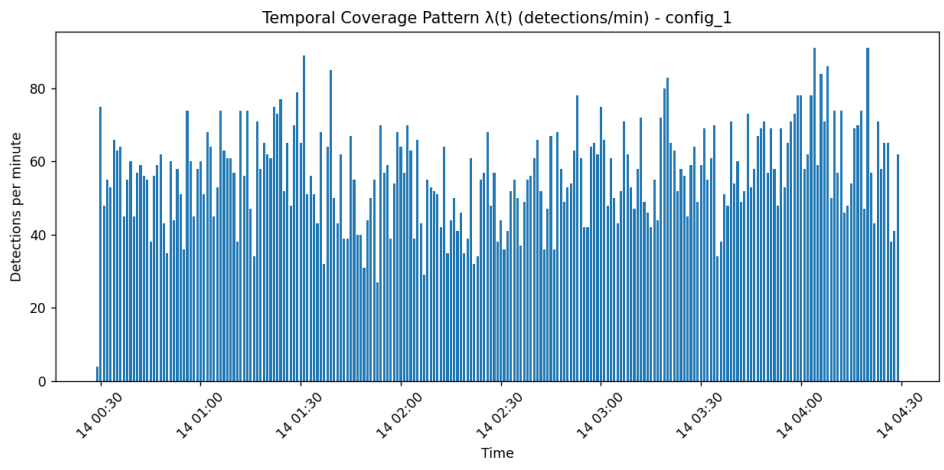


Figure 8: Time series of detections per minute/hour]

4.3 Category Analysis

4.3.1 Distribution of classes

Class counts are reported for each dataset/configuration. In uncontrolled data, emphasize category 69 (target) and −1 (noise/unknown).

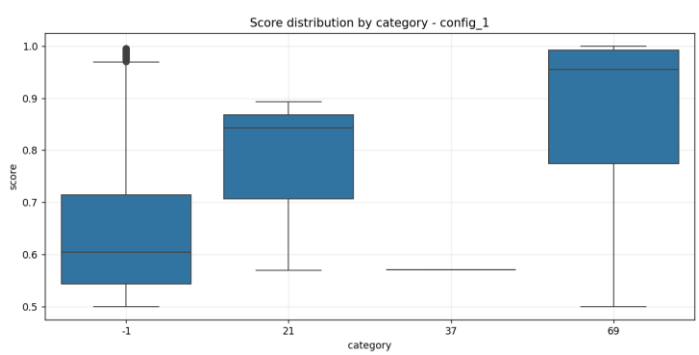
Config 1

```
Class distribution (category):
category
 69    11027
-1      2588
 21        3
 37        1
```

[Figure 9: Class distribution (bar plot)]

4.3.2 Score distribution by class

Use boxplots plots to compare score distributions across categories. Draft qualitative findings: category 69 shows high median score and low variance (distinct signature), while category −1 (noise) can have a wide score range up to 1.0, which can drive false positives.



[Figure 10: Score by category (boxplot)]

4.3.3 Signal quality metrics by category

Compute mean and standard deviation for score, signal_power, and SINR per category. Draft observation: category 21 exhibits high average SINR with low variance (stable strong signal), whereas category 69 may show higher variability due to changing environmental conditions.

Config 1

```
Mean metrics per category:
             score  signal_power       sinr
category
-1        0.645656   -115.007729   9.259066
 21       0.768582    -99.486100  21.354967
 37       0.571122    -92.135700  33.061900
 69       0.870634   -109.144736  15.421089
```

```
Std metrics per category:
             score  signal_power      sinr
category
-1        0.128125      6.226138  6.329295
 21       0.174118      5.052252  5.563151
 37            NaN           NaN       NaN
 69       0.154952      8.841467  9.656971
```

[Table 2: Per-category mean/std for score, power, SINR]

4.3.4 Miss detections per category (controlled)

In controlled data, miss detections are cases where GT label is a signal class but Pred Label is 0 (background noise). Report miss counts and miss rate per class and per RBW.

```
--- Miss Detections Analysis (Controlled) ---
Class 14.0: Miss Rate = 6.71%
Class 38.0: Miss Rate = 3.12%
Class 69.0: Miss Rate = 33.08%
```

[Figure 11: Miss detections by GT label and RBW]

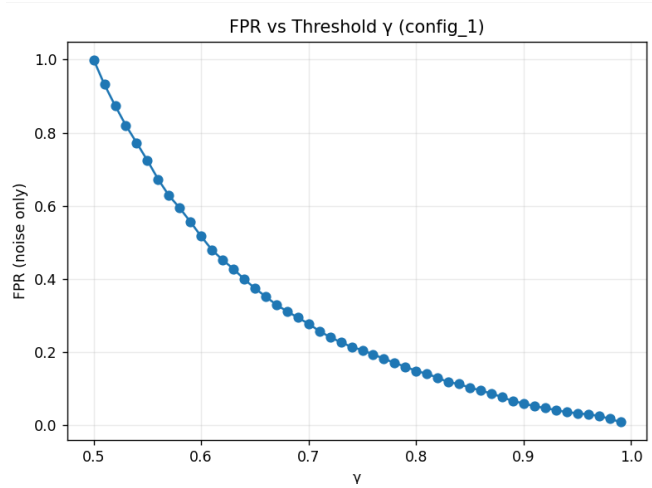
4.4 False Positive Rate (FPR) Analysis in Uncontrolled Data

Because uncontrolled data lacks complete ground truth over all time/frequency windows, true negatives (TN) are not directly observable. A practical proxy is to analyze how often noise-labeled detections (category = −1) exceed a decision threshold γ.

4.4.1 FPR as a function of score threshold

Compute pseudo-FPR(γ) by treating category −1 as negatives and counting events above threshold as FP, and those below as pseudo-TN.

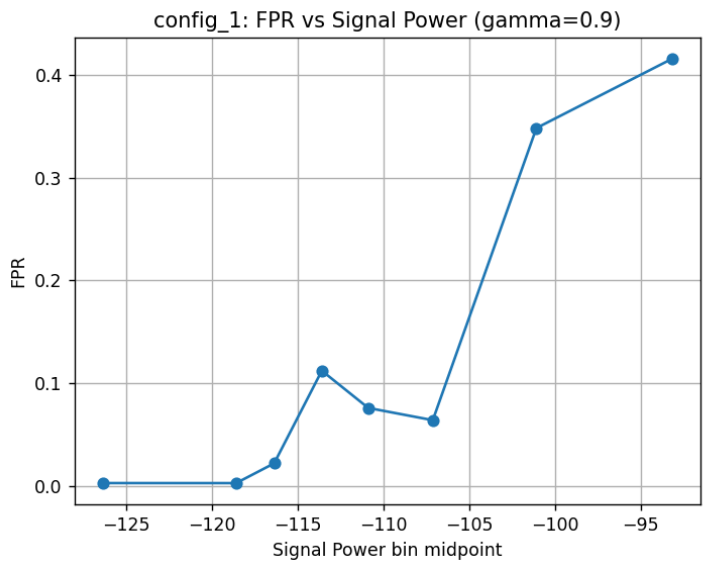
Draft observation: pseudo-FPR decreases monotonically as γ increases; at low γ (≈0.5) FPR is near 1, and for high γ (≈0.9–0.99) FPR approaches 0.



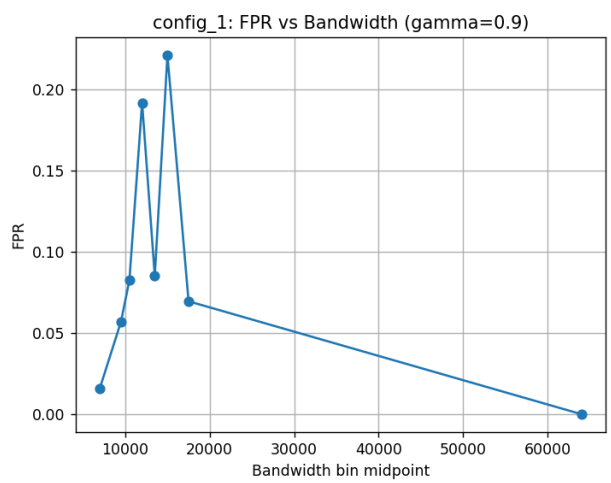
[Figure 12: pseudo-FPR vs score threshold γ]

4.4.2 FPR as a function of physical parameters

Repeat the pseudo-FPR analysis by conditioning on signal power and bandwidth bins. Draft observation: FPR rises sharply as power increases from about −105 dBm to −95 dBm, suggesting that stronger noise-like artifacts are misclassified. Bandwidth-dependent trends can indicate regimes where the detector confuses mid-bandwidth patterns with real signals.



[Figure 13: pseudo-FPR vs signal_power bins]



[Figure 14: pseudo-FPR vs bandwidth bins]

4.4.3 Reliability of FPR estimation

A fully reliable FPR cannot be estimated from the uncontrolled logs alone because the dataset contains only triggered detections, and therefore does not provide a complete measure of background-only windows required to count TN. A more reliable experiment would include (i) controlled injection windows to define true negatives, and/or (ii) external labeling (manual review or an independent sensor) to provide ground truth negatives over time.

4.5 TPR / Recall Analysis and Configuration Tradeoff

Absolute recall (TPR) can be computed from controlled data because the number of activated targets is known. Conversely, uncontrolled data provides the operational false-alarm behavior for a realistic 24/7 environment.

Cross-evaluation approach use controlled data to quantify Pd/recall as a function of γ, and uncontrolled data to quantify the operational cost Pfa/FPR. This aligns with a Neyman–Pearson perspective: choose a threshold to satisfy a false-alarm budget while maximizing detection power.

```
--- Precision/Recall per class (rbw=300Hz, gamma=0.9) ---
    class   TP  FN  FP    recall  precision
0      14   47   6   4  0.886792   0.921569
1      21   18   1   3  0.947368   0.857143
2      37   22   0   0  1.000000   1.000000
3      38   17   0   1  1.000000   0.944444
```

Figure 15: Controlled ROC/Recall vs threshold per RBW]

5. Task 2: Alert Trigger Design (Reducing False Positives)

Goal: reduce operational false positives while minimizing the impact on recall. Instead of alerting on every detection, an alert is raised only when additional criteria are satisfied.

5.1 Base threshold selection (Neyman–Pearson)

We apply the Neyman–Pearson (NP) principle to select a score threshold that satisfies a false-alarm budget. Let H_0 denote the noise hypothesis and let the detector score be the test statistic. For a given threshold γ, we estimate the false positive rate as:

$\mathrm{FPR}(\gamma) = P(\text{score} \geq \gamma \mid \text{noise})$.

Given an operational budget α (here $\alpha = 0.05$), the NP threshold is chosen as the smallest γ^* such that:

$\mathrm{FPR}(\gamma^*) \leq \alpha$.

In practice, we compute $\mathrm{FPR}(\gamma)$ over a grid of thresholds using noise-labeled detections (category = −1) and select the first threshold that meets the constraint. The resulting thresholds were $\gamma^* \approx 0.92$ for configuration 1 and $\gamma^* \approx 0.99$ for configuration 2, reflecting a stronger score overlap between signal and noise in configuration 2.

5.2 Additional reliability gate

Thresholding on score alone can still produce false alerts because occasional noise detections may achieve high confidence. Therefore, we apply lightweight reliability gates at the track level using association_id. Detections are grouped into tracks, and per-track features are computed (e.g., max score, number of hits, track duration, and median SINR/power). An alert is raised only if all conditions hold:

(1) Score gate (NP): $\text{max_score}(\text{track}) \geq \gamma^*$

(2) Persistence gate: $\text{n_hits}(\text{track}) \geq K$ OR $\text{duration}(\text{track}) \geq \Delta t$

(3) Quality gate (optional): $\text{median_SINR}(\text{track}) \geq \text{SINR_min}$ (and/or power constraints).

In our implementation, we used $K = 2$ hits, $\Delta t = 1$ second, and SINR_min = 3 dB. This track-based gating reduces spurious one-off detections while preserving most legitimate tracks.

5.3 Explainability and per-class performance

We report trigger performance at the track level because alerts are operationally generated per tracked signal (association_id). For each configuration we compare:

- Baseline: score-only trigger at $\gamma = 0.90$
- NP-only: score-only trigger at $\gamma = \gamma^*$
- Final trigger: NP threshold plus the track-level gates.

Configuration 1 results:

Baseline ($\gamma = 0.90$): FPR_track ≈ 0.060, RelativeRecall_track ≈ 0.601

NP-only ($\gamma^* \approx 0.92$): FPR_track ≈ 0.047, RelativeRecall_track ≈ 0.568

Final trigger (NP + gates): FPR_track ≈ 0.046, RelativeRecall_track ≈ 0.555

Configuration 2 results:

Baseline ($\gamma = 0.90$): FPR_track ≈ 0.110, RelativeRecall_track ≈ 0.288

NP-only ($\gamma^* \approx 0.99$): FPR_track ≈ 0.045, RelativeRecall_track ≈ 0.082

Final trigger (NP + gates): FPR_track ≈ 0.045, RelativeRecall_track ≈ 0.082

Interpretation: Configuration 1 achieves a meaningful reduction in false alerts with a moderate recall loss. In configuration 2, satisfying $\alpha = 0.05$ requires a very high γ^*, which strongly reduces recall; this indicates that additional feature engineering (frequency stability gating) or relaxing α may be necessary for a better operational tradeoff.

6. Conclusion

The controlled dataset enables reliable estimation of recall and miss-detection behavior per class and RBW. The uncontrolled datasets, while lacking full TN ground truth, provide valuable insight into operational false alarms driven primarily by category −1. A threshold chosen under a false-alarm budget, combined with a simple persistence or quality gate, is expected to substantially reduce false alerts with minimal recall loss.

APPENDIX

==========

TASK 1 — CONTROLLED DATASET

==========

Shape: (1161, 4)

Columns: ['GT labels', 'pred labels', 'pred scores', 'rbw']

Total detections: 1161

Unique GT labels (including 0): 13

Unique signals in controlled (excluding 0): 12

--- Miss Detections Analysis (Controlled) ---

Class 14.0: Miss Rate = 6.71%

Class 38.0: Miss Rate = 3.12%

Class 69.0: Miss Rate = 33.08%

--- Precision/Recall per class (rbw=2KHz, gamma=0.9) ---

	class	TP	FN	FP	recall	precision
0	14	91	5	6	0.947917	0.938144
1	21	5	6	0	0.454545	1.000000
2	37	9	8	0	0.529412	1.000000
3	38	14	1	0	0.933333	1.000000
4	68	12	0	0	1.000000	1.000000
5	69	358	262	3	0.577419	0.991690
6	70	1	0	0	1.000000	1.000000

7	94	1	0	0	1.000000	1.000000
8	95	0	1	0	0.000000	NaN
9	96	2	0	0	1.000000	1.000000
10	97	1	0	0	1.000000	1.000000
11	98	0	1	0	0.000000	NaN

--- Precision/Recall per class (rbw=300Hz, gamma=0.9) ---

	class	TP	FN	FP	recall	precision
0	14	47	6	4	0.886792	0.921569
1	21	18	1	3	0.947368	0.857143
2	37	22	0	0	1.000000	1.000000
3	38	17	0	1	1.000000	0.944444
4	68	4	1	0	0.800000	1.000000
5	69	132	52	3	0.717391	0.977778
6	70	2	0	0	1.000000	1.000000
7	94	4	0	0	1.000000	1.000000
8	95	5	0	0	1.000000	1.000000
9	96	5	0	0	1.000000	1.000000
10	97	1	0	0	1.000000	1.000000

==========

TASK 1 — UNCONTROLLED ANALYSIS: config_1

==========

Average duration: 34.60 sec

Mean signal_power: -110.26 | Median: -112.14

Mean SINR: 14.25 | Median: 11.28

Class distribution (category):

```
category
 69    11027
-1      2588
 21        3
 37        1
Name: count, dtype: int64

Mean metrics per category:
            score  signal_power       sinr
category
-1       0.645656   -115.007729   9.259066
 21      0.768582    -99.486100  21.354967
 37      0.571122    -92.135700  33.061900
 69      0.870634   -109.144736  15.421089

Std metrics per category:
            score  signal_power      sinr
category
-1       0.128125      6.226138  6.329295
 21      0.174118      5.052252  5.563151
 37           NaN           NaN       NaN
 69      0.154952      8.841467  9.656971

FPR curve (noise only) sample:
   gamma       FPR
0   0.50  0.999614
1   0.51  0.931994
2   0.52  0.874034
3   0.53  0.819938
4   0.54  0.772411
```

FPR vs signal_power bins (gamma=0.9)

	bin	N_noise	FPR
0	(-133.189, -119.453]	614	0.003257
1	(-119.453, -117.698]	622	0.003215
2	(-117.698, -115.024]	446	0.022422
3	(-115.024, -112.141]	258	0.112403
4	(-112.141, -109.61]	223	0.076233
5	(-109.61, -104.622]	202	0.064356
6	(-104.622, -97.537]	158	0.348101
7	(-97.537, -88.818]	65	0.415385

FPR vs bandwidth bins (gamma=0.9)

	bin	N_noise	FPR
0	(4999.999, 9000.0]	1442	0.015950
1	(9000.0, 10000.0]	422	0.056872
2	(10000.0, 11000.0]	182	0.082418
3	(11000.0, 13000.0]	261	0.191571
4	(13000.0, 14000.0]	47	0.085106
5	(14000.0, 16000.0]	154	0.220779
6	(16000.0, 19000.0]	72	0.069444
7	(19000.0, 109000.0]	8	0.000000

[Debug] bandwidth nunique (all): 46

bandwidth

13000.0	1534
11000.0	1526
15000.0	1442
17000.0	1293
19000.0	1192

```
10000.0    1128
9000.0      998
12000.0     940
8000.0      784
14000.0     776
Name: count, dtype: int64
[Debug] bandwidth nunique (noise): 21
bandwidth
8000.0     473
7000.0     432
10000.0    422
9000.0     420
11000.0    182
13000.0    153
15000.0    123
12000.0    108
6000.0      72
17000.0     52
Name: count, dtype: int64

--- Proxy Trigger Eval (event-level) gamma=0.9 ---
False alarms (noise events passing): 155
Estimated FPR (noise events): 0.0599

Proxy Relative Recall per class (uncontrolled):
   class  Pseudo_TP (Survived)  Pseudo_FN (Filtered)  RelativeRecall
0     21                     0                     3         0.00000
1     37                     0                     1         0.00000
2     69                  6628                  4399         0.60107
```

==========

TASK 1 — UNCONTROLLED ANALYSIS: config_2

==========

Average duration: 31.12 sec

Mean signal_power: -117.02 | Median: -120.02

Mean SINR: 12.99 | Median: 10.97

Class distribution (category):

```
category
 69    17039
-1     10761
 14     1268
 37      167
 21      153
 38       39
 68        6
Name: count, dtype: int64
```

Mean metrics per category:

category	score	signal_power	sinr
-1	0.648804	-120.104601	9.364726
14	0.579112	-120.810756	12.190596
21	0.543091	-113.782425	10.936880
37	0.565679	-108.314849	7.066383
38	0.551750	-101.823687	21.459656
68	0.656655	-107.819667	27.065885
69	0.767392	-114.937544	15.399451

Std metrics per category:

category	score	signal_power	sinr
-1	0.147719	6.517093	5.898439
14	0.083183	4.889339	3.704654
21	0.042669	5.598499	5.171846
37	0.055417	1.692311	2.867627
38	0.071704	3.870340	3.901349
68	0.142738	8.037691	12.470802
69	0.168543	9.310393	10.018648

FPR curve (noise only) sample:

	gamma	FPR
0	0.50	0.998885
1	0.51	0.917852
2	0.52	0.842951
3	0.53	0.777530
4	0.54	0.720007

FPR vs signal_power bins (gamma=0.9)

	bin	N_noise	FPR
0	(-138.435, -125.276]	1868	0.003747
1	(-125.276, -123.988]	2052	0.004873
2	(-123.988, -121.881]	1674	0.020908
3	(-121.881, -120.022]	1434	0.101813
4	(-120.022, -116.622]	1349	0.067457
5	(-116.622, -111.975]	1081	0.182239
6	(-111.975, -105.919]	862	0.520882
7	(-105.919, -89.224]	441	0.571429

FPR vs bandwidth bins (gamma=0.9)

	bin	N_noise	FPR
0	(-0.001, 3000.0]	3175	0.022992
1	(3000.0, 5000.0]	1835	0.055586
2	(5000.0, 8000.0]	2301	0.178183
3	(8000.0, 9000.0]	975	0.109744
4	(9000.0, 11000.0]	1437	0.200418
5	(11000.0, 12000.0]	291	0.261168
6	(12000.0, 14000.0]	325	0.264615
7	(14000.0, 50000.0]	422	0.106635

[Debug] bandwidth nunique (all): 39

bandwidth

10000.0 2659

9000.0 2647

11000.0 2592

8000.0 2533

12000.0 2487

13000.0 2404

1000.0 1999

4000.0 1871

3000.0 1770

7000.0 1571

Name: count, dtype: int64

[Debug] bandwidth nunique (noise): 33

bandwidth

3000.0 1266

4000.0 1172

8000.0 1168

```
1000.0     1069
9000.0      975
10000.0     917
7000.0      732
2000.0      690
5000.0      663
11000.0     520
Name: count, dtype: int64

--- Proxy Trigger Eval (event-level) gamma=0.9 ---
False alarms (noise events passing): 1187
Estimated FPR (noise events): 0.1103

Proxy Relative Recall per class (uncontrolled):
   class  Pseudo_TP (Survived)  Pseudo_FN (Filtered)  RelativeRecall
0     14                     7                  1261        0.005521
1     21                     0                   153        0.000000
2     37                     0                   167        0.000000
3     38                     0                    39        0.000000
4     68                     0                     6        0.000000
5     69                  5378                 11661        0.315629

==================================================================================
TASK 2 — TRIGGER DESIGN (NP + gates): config_1
==================================================================================
[NP] alpha=0.05 => gamma*=0.920

[Task2] FPR vs signal_power bins (gamma*=0.920)
```

	bin	N_noise	FPR
0	(-133.189, -119.453]	614	0.003257
1	(-119.453, -117.698]	622	0.001608
2	(-117.698, -115.024]	446	0.004484
3	(-115.024, -112.141]	258	0.077519
4	(-112.141, -109.61]	223	0.049327
5	(-109.61, -104.622]	202	0.049505
6	(-104.622, -97.537]	158	0.329114
7	(-97.537, -88.818]	65	0.369231

[Task2] FPR vs bandwidth bins (gamma*=0.920)

	bin	N_noise	FPR
0	(4999.999, 9000.0]	1442	0.008322
1	(9000.0, 10000.0]	422	0.035545
2	(10000.0, 11000.0]	182	0.065934
3	(11000.0, 13000.0]	261	0.176245
4	(13000.0, 14000.0]	47	0.085106
5	(14000.0, 16000.0]	154	0.188312
6	(16000.0, 19000.0]	72	0.055556
7	(19000.0, 109000.0]	8	0.000000

--- BASELINE (score-only) ---

gamma=0.9

N_tracks: 13619

N_noise_tracks: 2588

N_signal_tracks: 11031

Triggered_tracks: 6783

FPR_track: 0.05989180834621329

RelativeRecall_track: 0.6008521439579367

--- NP-ONLY (score-only) ---

gamma*=0.920

N_tracks: 13619

N_noise_tracks: 2588

N_signal_tracks: 11031

Triggered_tracks: 6393

FPR_track: 0.04714064914992272

RelativeRecall_track: 0.5684888042788505

--- FINAL TRIGGER (NP + gates) ---

gamma*=0.920, min_hits=2, min_dur=1.0, min_sinr=3.0, min_power=None

N_tracks: 13619

N_noise_tracks: 2588

N_signal_tracks: 11031

Triggered_tracks: 6243

FPR_track: 0.04559505409582689

RelativeRecall_track: 0.5552533768470673

--- SUMMARY (Before -> After) ---

FPR_track: 0.060 -> 0.046

RelativeRecall_track: 0.601 -> 0.555

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TASK 2 — TRIGGER DESIGN (NP + gates): config_2

==========

[NP] alpha=0.05 => gamma*=0.990

[Task2] FPR vs signal_power bins (gamma*=0.990)

	bin	N_noise	FPR
0	(-138.435, -125.276]	1868	0.000000
1	(-125.276, -123.988]	2052	0.000487
2	(-123.988, -121.881]	1674	0.000597
3	(-121.881, -120.022]	1434	0.009763
4	(-120.022, -116.622]	1349	0.027428
5	(-116.622, -111.975]	1081	0.077706
6	(-111.975, -105.919]	862	0.274942
7	(-105.919, -89.224]	441	0.240363

[Task2] FPR vs bandwidth bins (gamma*=0.990)

	bin	N_noise	FPR
0	(-0.001, 3000.0]	3175	0.000315
1	(3000.0, 5000.0]	1835	0.003815
2	(5000.0, 8000.0]	2301	0.095176
3	(8000.0, 9000.0]	975	0.044103
4	(9000.0, 11000.0]	1437	0.091858
5	(11000.0, 12000.0]	291	0.116838
6	(12000.0, 14000.0]	325	0.104615
7	(14000.0, 50000.0]	422	0.023697

--- BASELINE (score-only) ---

gamma=0.9

N_tracks: 29433

N_noise_tracks: 10761

N_signal_tracks: 18672

Triggered_tracks: 6572

FPR_track: 0.11030573366787473

RelativeRecall_track: 0.2883997429305913

--- NP-ONLY (score-only) ---

gamma*=0.990

N_tracks: 29433

N_noise_tracks: 10761

N_signal_tracks: 18672

Triggered_tracks: 2019

FPR_track: 0.04460551993309172

RelativeRecall_track: 0.0824228791773779

--- FINAL TRIGGER (NP + gates) ---

gamma*=0.990, min_hits=2, min_dur=1.0, min_sinr=3.0, min_power=None

N_tracks: 29433

N_noise_tracks: 10761

N_signal_tracks: 18672

Triggered_tracks: 2004

FPR_track: 0.04460551993309172

RelativeRecall_track: 0.08161953727506427

--- SUMMARY (Before -> After) ---

FPR_track: 0.110 -> 0.045

RelativeRecall_track: 0.288 -> 0.082